Obstacle-Avoiding Rover: Code

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Board: Arduino Uno (Nano works too, same pins). Serial Monitor: 9600 baud.
To drive the rover: battery switch off, upload over USB, unplug USB, battery switch on.
Pins: ENA D5, IN1 D7, IN2 D8, IN3 D9, IN4 D10, ENB D6, TRIG D12, ECHO D11, servo D3.

Step 1: Motor test

Lift the rover so the wheels spin in the air.

```
// STEP 1: Motor test - each motion runs for 2 seconds
// Lift the rover so the wheels spin in the air!
const uint8_t ENA = 5, IN1 = 7, IN2 = 8;    // left motor
const uint8_t ENB = 6, IN3 = 9, IN4 = 10;   // right motor
const uint8_t SPEED = 180;                  // 0 to 255

void setup() {
  pinMode(ENA, OUTPUT); pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);
  pinMode(ENB, OUTPUT); pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);
  delay(2000);                              // time to step back
}

void drive(bool lFwd, bool rFwd) {
  digitalWrite(IN1, lFwd);  digitalWrite(IN2, !lFwd);
  digitalWrite(IN3, rFwd);  digitalWrite(IN4, !rFwd);
  analogWrite(ENA, SPEED);  analogWrite(ENB, SPEED);
}

void stopMotors() {
  analogWrite(ENA, 0);  analogWrite(ENB, 0);
}

void loop() {
  drive(true, true);   delay(2000);   // forward
  stopMotors();        delay(1000);
  drive(false, false); delay(2000);   // backward
  stopMotors();        delay(1000);
  drive(false, true);  delay(2000);   // spin left
  stopMotors();        delay(1000);
  drive(true, false);  delay(2000);   // spin right
  stopMotors();        delay(1000);
}
```

Step 2: Ultrasonic sensor test

Open the Serial Monitor at 9600 baud. The battery switch can stay off.

```
// STEP 2: Distance sensor test
const uint8_t TRIG = 12, ECHO = 11;

void setup() {
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  Serial.begin(9600);
}

long readCm() {
  digitalWrite(TRIG, LOW);  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  long us = pulseIn(ECHO, HIGH, 30000);   // timeout 30 ms (about 5 m)
  if (us == 0) return 400;                // no echo: nothing detected
  return us / 58;
}

void loop() {
  Serial.print("Distance: ");
  Serial.print(readCm());
  Serial.println(" cm");
  delay(200);
}
```

Step 3: Servo test

Sensor on the servo horn; 90 degrees must point straight ahead.

```
// STEP 3: Servo test - sweep left, centre, right, centre
#include <Servo.h>

const uint8_t SERVO_PIN = 3;
Servo head;

void setup() {
  head.attach(SERVO_PIN);
  head.write(90);                 // centre
  delay(1000);
}

void loop() {
  head.write(160); delay(1000);   // should look LEFT
  head.write(90);  delay(1000);   // centre: sensor points straight ahead
  head.write(20);  delay(1000);   // should look RIGHT
  head.write(90);  delay(1000);   // centre
}
```

Step 4: Full obstacle-avoiding rover

Needs the Servo library (included with the Arduino IDE).

```
// STEP 4: Obstacle-avoiding rover
#include <Servo.h>

// ---- Pins ----
const uint8_t ENA = 5, IN1 = 7, IN2 = 8;      // left motor
const uint8_t ENB = 6, IN3 = 9, IN4 = 10;     // right motor
const uint8_t TRIG = 12, ECHO = 11;           // ultrasonic
const uint8_t SERVO_PIN = 3;

// ---- Tunable settings ----
const int  CRUISE_SPEED = 180;   // forward speed (0-255)
const int  TURN_SPEED   = 190;   // spinning speed
const int  STOP_CM      = 20;    // react when closer than this
const int  TURN_MS_90   = 450;   // time to turn about 90 degrees (calibrate!)

// Servo angles. If the rover turns TOWARD a wall instead of away
// from it, swap LOOK_LEFT and LOOK_RIGHT.
const int  LOOK_LEFT = 160, LOOK_CENTER = 90, LOOK_RIGHT = 20;

Servo head;

// ---------- Sensing ----------
long readCm() {
  digitalWrite(TRIG, LOW);  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  long us = pulseIn(ECHO, HIGH, 30000);
  if (us == 0) return 400;        // no echo: treated as "nothing in range"
  return us / 58;
}

// Median of 3 readings: one bad echo (too high OR too low) is ignored
long readCmSmooth() {
  long a = readCm(); delay(10);
  long b = readCm(); delay(10);
  long c = readCm();
  if (a > b) { long t = a; a = b; b = t; }
  if (b > c) { long t = b; b = c; c = t; }
  if (a > b) b = a;
  return b;                       // the middle value
}

long lookAt(int angle) {
  head.write(angle);
  delay(450);                     // give the servo time to arrive
  return readCmSmooth();
}

// ---------- Motion ----------
// speed: -255 (full reverse) to +255 (full forward)
```

```
51 void setMotor(uint8_t en, uint8_t inA, uint8_t inB, int speed) {
52   digitalWrite(inA, speed > 0);
53   digitalWrite(inB, speed < 0);
54   analogWrite(en, abs(speed));
55 }
56
57 void drive(int left, int right) {
58   setMotor(ENA, IN1, IN2, left);
59   setMotor(ENB, IN3, IN4, right);
60 }
61
62 void stopMotors()    { drive(0, 0); }
63 void forward()       { drive(CRUISE_SPEED, CRUISE_SPEED); }
64 void backward()      { drive(-CRUISE_SPEED, -CRUISE_SPEED); }
65 void spinLeft()      { drive(-TURN_SPEED, TURN_SPEED); }
66 void spinRight()     { drive(TURN_SPEED, -TURN_SPEED); }
67
68 // ---------- Setup ----------
69 void setup() {
70   pinMode(ENA, OUTPUT); pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);
71   pinMode(ENB, OUTPUT); pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);
72   pinMode(TRIG, OUTPUT); pinMode(ECHO, INPUT);
73
74   head.attach(SERVO_PIN);
75   head.write(LOOK_CENTER);
76   Serial.begin(9600);
77   delay(2000);                    // 2 s to put the rover on the floor
78 }
79
80 // ---------- Main loop ----------
81 void loop() {
82   long ahead = readCmSmooth();
83   Serial.print("Ahead: "); Serial.println(ahead);
84
85   if (ahead > STOP_CM) {
86     forward();
87     delay(50);
88     return;
89   }
90
91   // Obstacle detected
92   stopMotors();
93   delay(200);
94   backward();  delay(300);
95   stopMotors();
96
97   long left  = lookAt(LOOK_LEFT);
98   long right = lookAt(LOOK_RIGHT);
99   head.write(LOOK_CENTER);
100  delay(300);
101
102  Serial.print("Left: ");  Serial.print(left);
103  Serial.print("  Right: "); Serial.println(right);
104
105  if (left < STOP_CM && right < STOP_CM) {
106    spinRight();                  // dead end: turn around (about 180 degrees)
107    delay(TURN_MS_90 * 2);
108  } else if (left > right) {
109    spinLeft();
110    delay(TURN_MS_90);
111  } else {
112    spinRight();
113    delay(TURN_MS_90);
114  }
115  stopMotors();
116  delay(100);
117 }
```